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230. Multicenter Assessment of the Accuracy of MIC Results for Colistin with MicroScan Dried Gram Negative MIC Panels using CLSI Breakpoints

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Background. A multicenter study was performed to evaluate the accuracy of MIC results for colistin on a MicroScan Dried Gram Negative MIC (MSDGN) Panel when compared to results obtained with frozen broth microdilution panels prepared according to CLSI/EUCAST methodology. Note: Colistin is not available in all countries.

Methods. MSDGN panels were evaluated at three clinical sites (including Beckman Coulter) by comparing MIC values obtained using the MSDGN panels to MICs obtained utilizing a CLSI/EUCAST broth microdilution reference panel. The study included 407 clinical isolates tested using the turbidity and Prompt® methods of inoculation. MSDGN panels were incubated in the WalkAway System (35 ± 1°C) and read on the WalkAway System, the autoSCAN-4 instrument, and read visually at 16-20 hours. Frozen reference panels were prepared and inoculated according to CLSI/EUCAST Joint Working Group Recommendations for MIC determination of Colistin (Polymyxin E) and incubated for 16-20 hours and read visually. CLSI M100 ED32 breakpoints (μg/mL) used for interpretation of MIC results were: Enterobacterales ≤ 2 I, ≥ 4 R; *P. aeruginosa* ≤ 2 I, ≥ 4 R.

Results. Essential, categorical agreement, and categorical errors were calculated using MIC results from WalkAway reads for MSDGN panels in Table 1. All other read methods yielded similar results. Categorical agreement was not calculated for *Acinetobacter* spp. as MIC only is reported.

Table 1 - Results

Organism Group	WalkAway Essential Agreement* (EA) %		WalkAway Categorical Agreement* (CA) %		WalkAway Very Major Error** (VMJ) %		WalkAway Major Error** (MAJ) %	
	P	T	P	T	P	T	P	T
Enterobacterales	99.2 (258/260)	99.2 (258/260)	98.1 (255/260)	98.5 (256/260)	1.1 (1/93)	0 (0/93)	0.6 (1/167)	1.2 (2/167)
Acinetobacter spp.	100 (19/19)	89.5 (17/19)	-	-	-	-	-	-
P. aeruginosa	96.4 (53/55)	98.2 (54/55)	90.9 (50/55)	96.4 (53/55)	33.3 (1/3)	33.3 (1/3)	1.9 (1/52)	0 (0/52)
All Organisms***	98.0 (392/400)	97.3 (396/407)	96.8 (305/315)	98.1 (309/315)	2.1 (2/96)	1.0 (1/96)	0.9 (2/219)	0.9 (2/219)

P = Prompt inoculation method, T = Turbidity inoculation method
*Overall EA is calculated for all organisms and overall CA is calculated for all organisms with interpretive criteria.
**Calculation excluding 1 well errors
*** All organisms include *Acinetobacter* spp., *Aeromonas* spp., *B. cepacia* complex, *Enterobacterales*, Other non-*Enterobacterales*, *P. aeruginosa*, and *S. maltophilia* organism groups.

Conclusion. This multicenter study showed that colistin MIC results for Gram Negative bacteria obtained with the MSDGN panel correlate well with MICs obtained using frozen reference panels with CLSI interpretive criteria.

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